Skills Summary

Describing a Scatter Plot: Direction, Form, Strength

Use this reference while completing your worksheet or when studying.

Skill 1 — Direction: which way does it go?

Direction is one of three words: **positive** (y tends to rise as x rises), **negative** (y tends to fall), or **none**.

How to decide with evidence, not a glance: compare the y-value at the far left with the y-value at the far right, then check that the points in between agree. Left $<$ right and climbing throughout $\Rightarrow$ positive. Left $>$ right and falling throughout $\Rightarrow$ negative. If the highest and lowest y-values sit in the *middle* rather than at the ends, there is no direction.

Example: A plot's leftmost point is $(1, 12)$ and its rightmost is $(9, 30)$, with the points in between rising steadily. Name the direction.
Step 1. Direction is a statement about how y changes across the whole plot, so compare the two ends first and then confirm the middle agrees.
Step 2. The left end is at 12 and the right end at 30, and every point in between climbs — so the comparison is representative.
$\Rightarrow$ $\mathbf{12 < 30}$, so the direction is **positive**

Try it: A plot's leftmost point is $(2, 45)$ and its rightmost is $(20, 18)$, with the points falling steadily. Compare the two end values.

check: $\mathbf{45 > 18}$, negative

Skill 2 — Form: what shape do the points make?

Form is **linear** (the points follow a straight line) or **nonlinear** (they bend).

The test: walk the points left to right and ask whether they ever *reverse*. A line only ever goes one way. A single turning point — a lowest or highest value in the middle rather than at an end — means the form is nonlinear.

Check form *before* naming a direction. On a curve, "positive" and "negative" do not describe anything.

Example: A plot has points $(1, 20)$, $(2, 12)$, $(3, 8)$, $(4, 13)$, $(5, 24)$. Is the form linear or nonlinear?
Step 1. The quickest evidence for form is a turning point, so find the x-value carrying the smallest y and see whether it sits at an end or in the middle.
Step 2. The smallest y is 8, sitting at $x = 3$ — in the middle. The points fall to it and then rise away from it.
$\Rightarrow$ turning point at $x = \mathbf{3}$, so the form is **nonlinear**

Try it: A plot has points $(2, 40)$, $(4, 22)$, $(6, 14)$, $(8, 20)$, $(10, 38)$. At which x does it turn?

check: $x = \mathbf{6}$, nonlinear

Skill 3 — Strength: how tight is the band?

Strength is how close the points sit to the trend line: **strong** (a narrow band), **moderate**, or **weak** (a wide cloud).

Residual = actual y − predicted y. Small residuals mean a strong association. The **range** of the residuals (largest minus smallest) measures the width of the band, so a small range means strong.

Strength is not slope. A steep line can have a wide cloud around it. Slope says how fast y changes; strength says how reliably.

Example: Five points have residuals $1, -1, 0, 1, -1$ from their trend line. How strong is the association?
Step 1. Strength is about the width of the band the points occupy, so summarise the residuals by their range rather than judging the picture by eye.
Step 2. The largest residual is 1 and the smallest is -1, so the range is $1-(-1)$.
⇒ range = **2**, a narrow band: **strong**

Try it: Five points have residuals $7, -6, 8, -7, 6$. Find the range and say how strong the association is.

check: range = **15**, weak

(!) Watch out: an **outlier** — one point far off the pattern — makes a strong association look weak and drags the mean toward itself. Say so explicitly: “strong positive linear, with one outlier” beats both “strong” and “weak”.